

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12387667
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Wang; Cheng-Kuang et al.

Pixel circuit

Abstract

Pixel circuit includes light emitting element, driving transistor, capacitor, writing circuit, first transistor and second transistor. Light emitting element is coupled to first system voltage source. Driving transistor is coupled to first node, second node and second system voltage source. Capacitor and writing circuit are coupled to second node and third node. Writing circuit writes a data voltage and an initial voltage to both terminals of capacitor according to a control signal. First transistor is coupled between driving transistor and light emitting element, and is conducted in response to a driving signal. Second transistor is coupled to first node and third node, and is conducted in response to driving signal to change data voltage and initial voltage at both terminals of capacitor to conduct driving transistor to generate a driving current between two system voltage sources, flowing through driving transistor and first transistor, to light up light emitting element.

Inventors:	Wang; Cheng-Kuang (Hsinchu, TW), Lin; Syuan-Ying (Hsinchu, TW)
Applicant:	AUO Corporation (Hsinchu, TW)
Family ID:	92157858
Assignee:	AUO CORPORATION (Hsinchu, TW)
Appl. No.:	18/929685
Filed:	October 29, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20250201176 A1	Jun. 19, 2025

Foreign Application Priority Data

TW	112149331	Dec. 18, 2023
----	-----------	---------------

Publication Classification

Int. Cl.: G09G3/32 (20160101)

U.S. Cl.:

CPC G09G3/32 (20130101); G09G2300/0866 (20130101); G09G2330/023 (20130101)

Field of Classification Search

CPC: G09G (3/32); G09G (2300/0866); G09G (2330/023)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2019/0287462	12/2018	Zhou	N/A	N/A
2021/0375205	12/2020	Dong	N/A	G09G 3/3291
2022/0051619	12/2021	Shiau et al.	N/A	N/A
2023/0186834	12/2022	Liu	345/55	G09G 3/32
2024/0153442	12/2023	Li	N/A	G09G 3/3233
2024/0203361	12/2023	Meng	N/A	G09G 3/3275

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
I848658	12/2023	TW	N/A

Primary Examiner: Sasinowski; Andrew

Attorney, Agent or Firm: WPAT, PC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Taiwan Application Serial Number 112149331, filed Dec. 18, 2023, which is herein incorporated by reference in its entirety.

BACKGROUND

Field of Invention

(2) The present disclosure relates to a display device. More particularly, the present disclosure relates to a pixel circuit applied to transparent field sequential color (FSC) display panels.

Description of Related Art

(3) Conventional mini light light-emitting diode (mini LED) require large driving currents. A power supply voltage that generates the driving current is prone to current errors, resulting in different voltages for each of pixels, causing errors in output currents.

(4) In addition, in the conventional pixel circuit, when a micro-light emitting diode needs to output high brightness, a pixel circuit needs to generate a large current. When a large current flows through a path between two power supply voltages, the greater a number of transistors between two power supply voltages, the higher a power consumption between two power supply voltages. In

addition, when a system voltage source outputs the same system voltage to pixel circuits at different distances, a resistance of each of wirings increases as the distance increases, causing a current to drop more.

(5) For the foregoing reasons, there is a need for providing a pixel circuit to solve the above problems encountered in related art approaches.

SUMMARY

(6) One aspect of the present disclosure provides a pixel circuit. The pixel circuit includes a light emitting element, a driving transistor, a capacitor, a writing circuit, a first transistor and a second transistor. The light emitting element is coupled to a first system voltage source. The driving transistor is coupled to a first node, a second node and a second system voltage source. The capacitor is coupled to the second node to a third node. The writing circuit is coupled to the second node and the third node, and is configured to write a data voltage and an initial voltage to both terminals of the capacitor respectively according to a control signal at a first stage. The first transistor is coupled between the driving transistor and the light emitting element, and is conducted in response to a driving signal at a second stage. The second transistor is coupled to the first node and the third node, and is conducted in response to the driving signal at the second stage to change the data voltage and the initial voltage at both terminals of the capacitor so that the driving transistor is conducted to generate a driving current between the first system voltage source and the second system voltage source, flowing through the driving transistor and the first transistor to light up the light emitting element.

(7) Another aspect of the present disclosure provides a pixel circuit. The pixel circuit includes a first light emitting element, a driving transistor, a capacitor, a writing circuit, a first driving circuit, a second light emitting element and a second driving circuit. The first light emitting element is couple to a first system voltage source. The driving transistor is coupled to a first node, a second node and a second system voltage source. The capacitor is coupled to the second node and a third node. The writing circuit is coupled to the second node and the third node, and is configured to write an initial voltage and a first data voltage to both terminals of the capacitor respectively according to a control signal at a first stage terminal. The first driving circuit is coupled to the first node and the third node, and is conducted in response to a first driving signal at a second stage to change the initial voltage and the first data voltage at both terminals of the capacitor at the first stage, so as to conduct the driving transistor, to light up the first light emitting element. The second light emitting element is coupled to the first system voltage source. The second driving circuit is coupled to the first node and the third node. The writing circuit is configured to write the initial voltage and a second data voltage to both terminals of the capacitor according to the control signal at a third stage. The second driving circuit is conducted in response to a second driving signal at a fourth stage to change the initial voltage and the second data voltage at both terminals of the capacitor at the third stage so as to conduct the driving transistor to light up the second light emitting element.

(8) In view of the aforementioned shortcomings and deficiencies of the prior art, the present disclosure provides a pixel circuit. Through a design of a pixel circuit of the present disclosure, a voltage difference between system voltage sources of a pixel circuit can be reduced, thereby reducing power consumption.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:

(2) FIG. 1 depicts a schematic diagram of a pixel circuit according to some embodiments of the

present disclosure;

(3) FIG. 2 depicts a signal timing diagram of a pixel circuit at a display stage according to some embodiments of the present disclosure;

(4) FIG. 3 depicts a signal timing diagram of a pixel circuit at a detecting stage according to some embodiments of the present disclosure;

(5) FIG. 4 depicts a schematic diagram of data voltages received by different pixel circuits versus driving currents curves according to some embodiments of the present disclosure;

(6) FIG. 5 depicts a schematic diagram of a pixel circuit according to some embodiments of the present disclosure; and

(7) FIG. 6 depicts a signal timing diagram of a pixel circuit according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

(8) Reference will now be made in detail to the present embodiments of the invention, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the description to refer to the same or like parts.

(9) The terminology used herein is for the purpose of describing particular example embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

(10) Furthermore, it should be understood that the terms, “comprising”, “including”, “having”, “containing”, “involving” and the like, used herein are open-ended, that is, including but not limited to.

(11) The terms used in this specification and claims, unless otherwise stated, generally have their ordinary meanings in the art, within the context of the disclosure, and in the specific context where each term is used. Certain terms that are used to describe the disclosure are discussed below, or elsewhere in the specification, to provide additional guidance to the practitioner skilled in the art regarding the description of the disclosure.

(12) Conventional method of driving micro-light emitting diodes is to use a plurality of signal lines (such as data lines) in a display device to simultaneously input gray-scale voltages to sub-pixel circuits corresponding to three primary colors of light to achieve a target display screen of mixed colors. However, when a micro-light emitting diode needs to output high brightness, a pixel circuit needs to generate a large current. When a large current flows through a path between two power supply voltages, the greater a number of transistors between two power supply voltages, the higher a power consumption between two power supply voltages. Following paragraphs of the present disclosure will describe how to improve the aforementioned problems.

(13) FIG. 1 depicts a schematic diagram of a pixel circuit pixel circuit **100** according to some embodiments of the present disclosure. In one embodiment, the pixel circuit **100** includes a light emitting element **L1**, a driving transistor **DT1**, a capacitor **C1**, a writing circuit **110**, a transistor **T1**, a transistor **T2**, a bypass transistor **BT1** and a detecting transistor **AT1**. Please start from a top side and a right side of each of components in the picture as a first terminal, the light emitting element **L1** includes a first terminal and a second terminal. The second terminal of the light emitting element **L1** is coupled to a system voltage source **VSS**. The transistor **T1** is coupled between the light emitting element **L1** and the driving transistor **DT1**.

(14) The driving transistor **DT1** includes a first terminal, a second terminal and a control terminal. The first terminal of the driving transistor **DT1** is coupled to the node **N1** and a system voltage source **VDD**. The second terminal of the driving transistor **DT1** is coupled to the transistor **T1**. The control terminal of the driving transistor **DT1** is coupled to the node **N2**.

(15) The transistor **T1** includes a first terminal, a second terminal and a control terminal. The first terminal of the transistor **T1** is coupled to the second terminal of the driving transistor **DT1**. The second terminal of the transistor **T1** is coupled to the first terminal of the light emitting element **L1**.

The control terminal of the transistor **T1** is configured to receive a driving signal $EM[n]$. The transistor **T1** is conducted in response to the driving signal $EM[n]$.

(16) The transistor **T2** includes a first terminal, a second terminal and a control terminal. The first terminal of the transistor **T2** is coupled to the node **N1**, the system voltage source VDD and the first terminal of the driving transistor **DT1**. The second terminal of the transistor **T2** is coupled to the node **N3**. The control terminal of the transistor **T2** is configured to receive the driving signal $EM[n]$. The transistor **T2** is conducted in response to the driving signal $EM[n]$.

(17) The capacitor **C1** includes a first terminal and a second terminal. The capacitor **C1** is coupled to the node **N2** and the node **N3**. The first terminal of the capacitor **C1** is coupled to the node **N3**. The second terminal of the capacitor **C1** is coupled to the node **N2**.

(18) Please continue to refer to FIG. 1, the writing circuit **110** is coupled to the node **N2** and the node **N3**. The writing circuit **110** includes a transistor **T3** and a transistor **T4**. The transistor **T3** includes a first terminal, a second terminal and a control terminal. The first terminal of the transistor **T3** is coupled to an initial voltage source V_{in} , and is configured to receive an initial voltage from the initial voltage source V_{in} . The second terminal of the transistor **T3** is coupled to the capacitor **C1** and the node **N3**. The control terminal of the transistor **T3** is configured to receive a control signal $SN[n]$. The transistor **T3** is conducted in response to the control signal $SN[n]$.

(19) The transistor **T4** includes a first terminal, a second terminal and a control terminal. The first terminal of the transistor **T4** is coupled to the capacitor **C1**. The second terminal of the transistor **T4** is configured to receive a data voltage V_{data} from a data line (not shown in the figure). The control terminal of the transistor **T4** is configured to receive a control signal $SN[n]$. The control terminal of the transistor **T4** is conducted in response to the control signal $SN[n]$.

(20) The bypass transistor **BT1** includes a first terminal, a second terminal and a control terminal. The first terminal of the bypass transistor **BT1** is coupled to the detecting transistor **AT1**. The second terminal of the bypass transistor **BT1** is coupled to the light emitting element **L1**. The control terminal of the bypass transistor **BT1** is configured to receive the driving signal $EM[n]$. The bypass transistor **BT1** is conducted in response to the driving signal $EM[n]$.

(21) The detecting transistor **AT1** includes a first terminal, a second terminal and a control terminal. The second terminal of the detecting transistor **AT1** is coupled to the first terminal of the bypass transistor **BT1**. The control terminal of the detecting transistor **AT1** is configured to receive a detection control signal AT . The detecting transistor **AT1** is conducted in response to the detection control signal AT .

(22) In one embodiment, the driving transistor **DT1**, the transistors **T1** to **T4**, bypass transistor **BT1** and the detecting transistor **AT1** can be implemented as P-type Metal-Oxide-Semiconductor Field-Effect Transistor (PMOS). However, types of the aforementioned transistors can be design according to actual needs and are not limited to the embodiments of the present disclosure.

(23) In some embodiments, in order to facilitate the understanding operation of the pixel circuit **100** in FIG. 1, please refer to FIG. 2. FIG. 2 depicts a signal timing diagram of the pixel circuit **100** in FIG. 1 at a display stage **I1** according to some embodiments of the present disclosure. The display stage **I1** consists of stages **I11** to **I14**.

(24) In some embodiments, please refer to FIG. 1 and FIG. 2, at the stage **I11** of the display stage **I1**, the control signal $SN[n]$ is at a low level L . The driving signal $EM[n]$ is at a high level H . The control signal $SN[n]$ writes the initial voltage of the initial voltage source V_{in} to the first terminal of the capacitor **C1** (i.e. the node **N3**) through the transistor **T3** of the writing circuit **110**. The control signal $SN[n]$ writes the data voltage V_{data} to the second terminal of the capacitor **C1** (i.e. the node **N2**) from the data line (not shown in the figure) through the transistor **T4** of the writing circuit **110**. At this time, a voltage level of the node **N2** changes as the voltage level $V_{sub.N2}$ shown in FIG. 2. A voltage level of the node **N3** changes as the voltage level $V_{sub.N3}$ shown in FIG. 2.

(25) For example, the initial voltage of the initial voltage source V_{in} written by the writing circuit

110 to the node **N3** is 10V. The data voltage V_{data} of the data line written by the writing circuit **110** to the node **N2** is 4V. At this time, the voltage level $V_{sub.N3}$ is 10V. The voltage level $V_{sub.N2}$ is 4V. The capacitor **C1** is configured to store a voltage difference of the node **N2** and the node **N3**. It should be noted that since the data voltage V_{data} is greater than a threshold voltage of the driving transistor **DT1**, and the driving transistor **DT1** is turned off at this time.

(26) Please refer to FIG. 1 and FIG. 2, at the stage **I12** of the display stage **I1**, the control signal $SN[n]$ and the driving signal $EM[n]$ are both at the high level H. At this time, the driving transistor **DT1**, the transistors **T1** to **T4**, the bypass transistor **BT1** and the detecting transistor **AT1** are all in a turned off state, to maintain the voltage level $V_{sub.N2}$ and the voltage level $V_{sub.N3}$ at both terminals of the capacitor **C1** at the stage **I11**. It should be noted that a purpose of setting stage **I12** is to prevent a signal delay of the control signal $SN[n]$ from affecting the following circuit operation of the pixel circuit **100**. In other words, when the control signal $SN[n]$ changes from the low level L to the high level H, the stage **I12** is configured to prevent the transistor **T3** and the transistor **T4** of the writing circuit **110** from being completely turned off, resulting in short circuit and other abnormal situations in a circuit.

(27) At the stage **I13** of the display stage **I1**, the driving signal $EM[n]$ is at the low level L. The control signal $SN[n]$ is at the high level H. The driving signal $EM[n]$ changes the voltage level $V_{sub.N3}$ of the node **N3** through the transistor **T2** and the system voltage of the system voltage source **VDD**. At this time, the capacitor **C1** changes the voltage level $V_{sub.N2}$ of the node **N2** in response to the voltage level $V_{sub.N3}$ of the node **N3**, thereby making the voltage level $V_{sub.N2}$ meet the turn-on condition of the driving transistor **DT1**. For example, the system voltage of the system voltage source **VDD** is 7.6V. The voltage level $V_{sub.N3}$ of the node **N3** drops from 10V to 7.6V. At this time, the capacitor **C1** changes the voltage level $V_{sub.N2}$ of the node **N2** in response to the change of the voltage level $V_{sub.N3}$ of the node **N3**. The voltage level $V_{sub.N2}$ of the node **N2** drops from 4V to 1.6V. The driving transistor **DT1** is conducted in response to the low voltage of the voltage level $V_{sub.N2}$ of the node **N2**.

(28) Then, after the driving transistor **DT1** is conducted, the driving signal $EM[n]$ generates a driving current I_{d1} between the system voltage source **VDD** and system voltage source **VSS** through the transistor **T1** and the driving transistor **DT1**, flowing through transistor **T1** and driving transistor **DT1** to light up the light emitting element **L1**. For example, the driving current I_{d1} is 50 microamperes.

(29) It should be noted that, please refer to FIG. 1, a path between the system voltage source **VDD** and the system voltage source **VSS** of the conventional pixel circuit has at least three transistors and light emitting elements, and conventional pixel circuit requires the system voltage source **VDD** to provide a voltage of 9V to generate the driving current I_{d1} of 50 microamperes (μA) driving current I_{d1} . In contrast, the pixel circuit **100** of the present disclosure only requires two transistors (i.e. the driving transistor **DT1** and the transistor **T1**) and the light emitting element **L1**, and only requires the system voltage source **VDD** to provide a voltage of 7.6V to generate the driving current I_{d1} of 50 microamperes. Under the same current value, the design of the pixel circuit **100** of the present disclosure saves about 16% of power. In addition, the pixel circuit **100** of the present disclosure can also enable the system voltage source **VDD** to provide a voltage of 9V to generate a higher current value of the driving current I_{d1} . It should be further noted that the current value of the aforementioned driving current I_{d1} and the voltage value of the system voltage can be design according to actual needs and are not limited to the embodiments of the present disclosure.

(30) At the stage **I14** of the display stage **I1**, the control signal $SN[n]$ and the driving signal $EM[n]$ are at the high level H. At this time, the driving transistor **DT1**, the transistors **T1** to **T4**, the bypass transistor **BT1** and the detecting transistor **AT1** are all in a turned off state. It should be noted that a setting purpose of the stage **I14** is similar to a setting purpose of the stage **I12**, and repetitious detailed descriptions are omitted here.

(31) FIG. 3 depicts a signal timing diagram of the pixel circuit **100** in FIG. 1 at a detecting stage

DI1 according to some embodiments of the present disclosure. The detecting DI1 consists of the stages DI11 to DI14. Operations of the pixel circuit 100 at the stages DI11 to DI14 of the detecting stage DI1 are respectively similar to the operations of the pixel circuit 100 at the stages I11 to I14 of the display stage I1. For the sake of brevity, only the differences are described below.

(32) Please refer to FIG. 1 to FIG. 3, at the stage DI13 of the detecting stage DI1, the detection control signal AT is at the low level VGL. The driving signal EM[n] is at the low level L. The detection control signal AT and the driving signal EM[n] bypass the driving current Id1 to the data line (not shown in the figure) through the bypass transistor BT1 and the detecting transistor AT1 respectively, so as to determine whether the pixel circuit 100 is operating normally through a waveform and a value of the driving current Id1.

(33) There are two differences between the stage I13 of the display stage I1 and the stage DI13 of the detecting stage DI1. A first difference is that the detection control signal AT has different levels. In detail, the detection control signal AT at the stage I13 of the display stage I1 is at a high level VGH. The detection control signal AT at the stage DI13 of the detecting stage DI1 is at a low level VGL. A second difference is that a path through which the driving current Id1 flows is different. In detail, the driving current Id1 of the stage I13 of the display stage I1 flows the driving transistor DT1, the transistor T1 and the light emitting element L1 in sequence. The driving current Id1 of the stage DI13 of the detecting stage DI1 flows the driving transistor DT1, the transistor T1, the bypass transistor BT1 and the detecting transistor AT1 to data line (not shown in the figure).

(34) In addition, when a voltage source of a conventional pixel device outputs the same voltage to pixel circuits with different distances, a resistance of each of wirings increase as distance increase, causing a current to drop more. Through a design of the pixel circuit 100 in FIG. 1 of the present disclosure, when a voltage source of a conventional pixel device outputs the same voltage to pixel circuits 100 with different distances, differences in driving current Id1 between pixel circuits 100 with different distances will also obtain a certain compensation effect.

(35) FIG. 4 depicts a schematic diagram of a data voltage received by different pixel circuits versus a driving current curves P1-P4 according to some embodiments of the present disclosure.

Horizontal axis coordinates of the voltage versus current curves P1-P4 represent the data voltage V data written by the pixel circuit 100, and its unit is volts (V). Vertical axis coordinates of the voltage versus current curves P1~P4 represent the driving current Id1 between the system voltage source VDD and the system voltage source VSS in the pixel circuit 100, and its unit is Ampere (A). The voltage versus current curve P1 is a voltage versus current curve of conventional pixel circuit that is close to a voltage source of the conventional pixel circuit. The voltage versus current curve P2 is a voltage versus current curve of conventional pixel circuit that is far away from the voltage source of the conventional pixel circuit. The voltage versus current curve P3 is a voltage versus current curve of the pixel circuit 100 that is close to a voltage source of the present disclosure. The voltage versus current curve P4 is a voltage versus current curve of the pixel circuit 100 that is far away from the voltage source of the present disclosure.

(36) Please refer to FIG. 1, a path between the system voltage source VDD and the system voltage source VSS of the conventional pixel circuit has at least three transistors and light emitting elements. Please refer to FIG. 4, conventional pixel circuit is configured to receive the same data voltage (e.g. 3V or 5V). Since resistances of wirings increase with distance, a gap between the voltage versus current curve P1 and the voltage versus current curve P2 is very large.

(37) Please refer to FIG. 1 again, a path between the system voltage source VDD and the system voltage source VSS of the pixel circuit 100 only has the driving transistor DT1, the transistor T1 and the light emitting element L1. Please refer to FIG. 4 again, through the transistor T2 of the pixel circuit 100 and the stage I13 in FIG. 2 of the present disclosure, a difference between the voltage versus current curves P3-P4 is narrowed compared to the difference between the voltage versus current curves P1-P2 to produce a compensation effect for the driving current Id1 in the farther pixel circuit 100.

(38) Conventional driving circuits that drive micro-light-emitting diodes with different colors of light also occupy an area of a display device. Following paragraphs of the present disclosure will further improve an area of a display device according to the design of the pixel circuit **100** of the present disclosure.

(39) FIG. 5 depicts a schematic diagram of the pixel circuit **100A** according to some embodiments of the present disclosure. The pixel circuit **100A** includes a light emitting element **L1**, a light emitting element **L2**, a light emitting element **L3**, a driving transistor **DT1**, a capacitor **C1**, a writing circuit **110A**, a driving circuit **120A**, a driving circuit **130A** and a driving circuit **140A**. The driving circuit **120A** includes a transistor **T1** and a transistor **T2**. The writing circuit **110A** includes a transistor **T3** and a transistor **T4**. The driving circuit **130A** includes a transistor **T5** and a transistor **T6**. The driving circuit **140A** includes a transistor **T7** and a transistor **T8**. The pixel circuit **100A** further includes a detecting transistor **AT1**, bypass transistors **BT1** to **BT3**. Connections between the light emitting element **L1**, the driving transistor **DT1**, the capacitor **C1**, the transistor **T1** and transistor **T2** of the driving circuit **120A**, the transistor **T3** and the transistor **T4** of the writing circuit **110A**, the bypass transistor **BT1** and the detecting transistor **AT1** are similar to the corresponding components of the pixel circuit **100** in FIG. 1, and repetitious detailed descriptions are omitted here. Following paragraphs will describe the structural differences between the pixel circuit **100** in FIG. 1 and the pixel circuit **100A** in FIG. 5.

(40) The driving circuit **130A** and the driving circuit **140A** are coupled to the node **N1** and the node **N3** like the driving circuit **120A**. The driving circuit **120A**, the driving circuit **130A** and the driving circuit **140A** share the driving transistor **DT1**, the capacitor **C1** and the writing circuit **110A**. Compared with conventional three-color sub-pixel circuit, this circuit sharing design reduces the area of the pixel circuit **100A** to increase an aperture ratio of a transparent display device.

(41) The transistor **T1** of the driving circuit **120A** is coupled to the light emitting element **L1** and the driving transistor **DT1**. The transistor **T6** of the driving circuit **130A** is coupled to the light emitting element **L2** and the driving transistor **DT1**. The transistor **T8** of the driving circuit **140A** is coupled to the light emitting element **L3** and the driving transistor **DT1**.

(42) The transistor **T5** and the transistor **T7** are connected in parallel to the transistor **T2**. The transistor **T6** and the bypass transistor **BT2** are connected in series. The transistor **T6** and the bypass transistor **BT2** are connected in parallel to the transistor **T1** and the bypass transistor **BT1**. The transistor **T8** and the bypass transistor **BT3** are connected in series. The transistor **T8** and the bypass transistor **BT3** are connected in parallel to the transistor **T2** and the bypass transistor **BT2**.

(43) The transistor **T5** includes a first terminal, a second terminal and a control terminal. The first terminal of the transistor **T5** is coupled to the node **N1**. The second terminal of the transistor **T5** is coupled to the node **N3**. The control terminal of the transistor **T5** is configured to receive a driving signal **EM2[n]**. The transistor **T5** is conducted in response to the driving signal **EM2[n]**. The transistor **T6** includes a first terminal, a second terminal and a control terminal. The first terminal of the transistor **T6** is coupled to the second terminal of the driving transistor **DT1**. The second terminal of the transistor **T6** is coupled to the light emitting element **L2**. The control terminal of the transistor **T6** is configured to receive the driving signal **EM2[n]**. The transistor **T6** is conducted in response to the driving signal **EM2[n]**.

(44) The transistor **T7** includes a first terminal, a second terminal and a control terminal. The first terminal of the transistor **T7** is coupled to the node **N1**. The second terminal of the transistor **T7** is coupled to the node **N3**. The control terminal of the transistor **T7** is configured to receive a driving signal **EM3[n]**. The transistor **T7** is conducted in response to the driving signal **EM3[n]**. The transistor **T8** includes a first terminal, a second terminal and a control terminal. The first terminal of the transistor **T8** is coupled to the second terminal of the driving transistor **DT1**. The second terminal of the transistor **T8** is coupled to the light emitting element **L3**. The control terminal of the transistor **T8** is configured to receive the driving signal **EM3[n]**. The transistor **T8** is conducted in response to the driving signal **EM3[n]**.

(45) The bypass transistor BT2 includes a first terminal, a second terminal and a control terminal. The first terminal of the bypass transistor BT2 is coupled to the second terminal of the detecting transistor AT1. The second terminal of the bypass transistor BT2s coupled to the light emitting element L2. The control terminal of the bypass transistor BT2 is configured to receive the driving signal EM2[n]. The bypass transistor BT2 is conducted in response to the driving signal EM2[n]. The bypass transistor BT3 includes a first terminal, a second terminal and a control terminal. The first terminal of the bypass transistor BT3 is coupled to the second terminal of the detecting transistor AT1. The second terminal of the bypass transistor BT3 is coupled to the light emitting element L3. The control terminal of the bypass transistor BT3 is configured to receive the driving signal EM3[n]. The bypass transistor BT3 is conducted in response to the driving signal EM3[n]. It should be noted that the detection control signal AT received by the control terminal of the detecting transistor AT1 is only at a low level (e.g. the low level VGL in FIG. 3) during the detecting stage. At the display stages I1-I3, the detection control signal AT is at a high level (e.g. the high level VGH in FIG. 2), whose operation is similar to the operation of the detecting stage DI1 in FIG. 3, and repetitious detailed descriptions are omitted here.

(46) In one embodiment, an optical wavelength of each of the light emitting element L1, the light emitting element L2 and the light emitting element L3 is different. For example, the light emitting element L1 can be a micro light-emitting diode (micro-LED) with a red light wavelength. The light emitting element L2 can be a micro light-emitting diode (micro-LED) with a green light wavelength. The light emitting element L3 can be a micro light-emitting diode (micro-LED) with a blue light wavelength. The light emitting element L1, the light emitting element L2 and the light emitting element L3 can be adjusted according to actual needs and are not limited to the embodiment of the present disclosure.

(47) In order to facilitate the understanding operation of the pixel circuit 100A in FIG. 5, please refer to FIG. 6 together. FIG. 6 depicts a signal timing diagram of the pixel circuit 100A in FIG. 5 at the display stages I1 to I3 according to some embodiments of the present disclosure. The display stage I1 consists of stages I11 to I14. The display stage I2 consists of stages I21 to I24. The display stage I3 consists of stages stage I31 to I34. Operations of the display stages I1 to I3 in FIG. 6 are similar to the display stage I1 in FIG. 2. For the sake of brevity, only the differences are described below.

(48) Please refer to the stage I11 of the display stage I1 in FIG. 6, the control signal SN[n] is at the low level L. The driving signal EM1[n], the driving signal EM2[n] and the driving signal EM3[n] are all at the high level H. The control signal SN[n] writes the initial voltage of the initial voltage source Vin to the first terminal (i.e. the node N3) of the capacitor C1 through the transistor T3 of the writing circuit 110A. The control signal SN[n] writes the data voltage Vdata from data line (not shown in the figure) to the second terminal (i.e. the node N2) of the capacitor C1 through the transistor T4 of the writing circuit 110A.

(49) Please refer to the stage I13 of the display stage I1 in FIG. 6, the driving signal EM1[n] is at the low level L. The control signal SN[n], the driving signal EM2[n] and the driving signal EM3[n] is at the high level H. The transistor T1 and the transistor T2 of the driving circuit 120A are conducted in response to the driving signal EM1[n] to change the voltage level V.sub.N2 of the node N2 and the voltage level V.sub.N3 of the node N3 at the stage I11 so as to conduct the driving transistor DT1 to generate the driving current Id1 between the system voltage source VDD and the system voltage source VSS, flowing through the transistor T1 and the driving transistor DT1 to light up the light emitting element L1. Detail content of operations is similar to the stage I13 in FIG. 2, and repetitious detailed descriptions are omitted here.

(50) Please refer to the stage I21 of the display stage I2 in FIG. 6, the control signal SN[n] is at the low level L. The driving signal EM1[n], the driving signal EM2[n] and the driving signal EM3[n] are at the high level H. The control signal SN[n] writes the initial voltage of the initial voltage source Vin to the first terminal (i.e. the node N3) the capacitor C1 through transistor T3 of the

writing circuit **110A**. The control signal SN[n] writes the data voltage Vdata from a data line (not shown in the figure) to the second terminal (i.e. the node N2) of the capacitor C1 through the transistor T4 of the writing circuit **110A**. The data voltage Vdata can be inputted different voltage levels or the same voltage level according to actual needs.

(51) Please refer to the stage I11 and the stage I21 in FIG. 6, the voltage level V2 of the data voltage Vdata at the stage I21 is higher than the voltage level V1 of the data voltage Vdata at the stage I11.

(52) Please refer to the stage I23 of the display stage I2 in FIG. 6, the driving signal EM2[n] is at the low level L. The control signal SN[n], the driving signal EM1[n] and the driving signal EM3[n] are at the high level H. The transistor T5 and the transistor T6 of the driving circuit **130A** are conducted in response to the driving signal EM2[n] to change the voltage level V.sub.N2 of the node N2 and the voltage level V.sub.N3 of the node N3 at the stage I21 so as to conduct the driving transistor DT1 to generate the driving current Id2 between the system voltage source VDD and the system voltage source VSS, flowing through the transistor T6 and the driving transistor DT1 to light up the light emitting element L2. Detail content of operations is similar to the stage I13 in FIG. 2, and repetitious detailed descriptions are omitted here.

(53) Please refer to stage I31 of the display stage I3 in FIG. 6, the control signal SN[n] is at the level L. The driving signal EM1[n], the driving signal EM2[n] and the driving signal EM3[n] at the high level H. The control signal SN[n] writes the initial voltage of the initial voltage source Vin to the first terminal (i.e. the node N3) of the capacitor C1 through the transistor T3 of the writing circuit **110A**. The control signal SN[n] writes the data voltage Vdata from the data line (not shown in the figure) to the second terminal (i.e. the node N2) of the capacitor C1 through the transistor T4 of the writing circuit **110A**. The data voltage Vdata can be inputted different voltage levels or the same voltage level according to actual needs.

(54) Please refer to the stage I11, the stage I21 and the stage I31 in FIG. 6, the voltage level V3 of the data voltage Vdata at the stage I31 is higher than the voltage level V1 of the data voltage Vdata at the stage I11 and the voltage level V2 of the data voltage Vdata at the stage I21.

(55) Please refer to the stage I33 of the display stage I3 in FIG. 6, the driving signal EM3[n] is at the low level L. The control signal SN[n], the driving signal EM1[n] and the driving signal EM2[n] are at the high level H. The transistor T7 and the transistor T8 of the driving circuit **140A** is conducted in response to the driving signal EM3[n] to change the voltage level V.sub.N2 of the node N2 and the voltage level V.sub.N3 of the node N3 at the stage I31 so as to conduct the driving transistor DT1 to generate the driving current Id3 between the system voltage source VDD and the system voltage source VSS, flowing through the transistor T8 and the driving transistor DT1 to light up the light emitting element L3. Detail content of operations is similar to the stage I13 in FIG. 2, and repetitious detailed descriptions are omitted here.

(56) To sum up, in addition to the advantages of the pixel circuit **100** in FIG. 1 the pixel circuit **100A** in FIG. 5 allows micro-light emitting diodes of multiple colors to share the same driving circuit, thereby reducing the area of the pixel circuit and increasing an aperture ratio of a pixel device.

(57) Based on the aforementioned embodiments, the present disclosure provides a pixel circuit to reduce a voltage difference between system voltage sources of a pixel circuit, thereby reducing power consumption. In addition, differences in driving currents between pixel circuits with different distances will also obtain a certain compensation effect. Finally, shared design of pixel circuits of the present disclosure can reduce an area of a pixel circuit and increase an aperture ratio of a pixel device.

(58) Although the present disclosure has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

(59) It will be apparent to those skilled in the art that various modifications and variations can be

made to the structure of the present disclosure without departing from the scope or spirit of the present disclosure. In view of the foregoing, it is intended that the present disclosure cover modifications and variations of the present disclosure provided they fall within the scope of the following claims.

Claims

1. A pixel circuit, comprising: a light emitting element, coupled to a first system voltage source; a driving transistor, coupled to a first node, a second node and a second system voltage source; a capacitor, coupled to the second node and a third node; a writing circuit, coupled to the second node and the third node, wherein the writing circuit is configured to write a data voltage and an initial voltage to both terminals of the capacitor respectively according to a control signal at a first stage; a first transistor, coupled between the driving transistor and the light emitting element, wherein the first transistor is conducted in response to a driving signal at a second stage; and a second transistor, coupled to the first node and the third node, wherein the second transistor is conducted in response to the driving signal at the second stage to change the data voltage and the initial voltage at both terminals of the capacitor so that the driving transistor is conducted to generate a driving current between the first system voltage source and the second system voltage source, flowing through the driving transistor and the first transistor to light up the light emitting element.
2. The pixel circuit of claim 1, wherein the writing circuit comprises: a third transistor, comprising: a first terminal, coupled to an initial voltage source, and configured to receive the initial voltage from the initial voltage source at the first stage; a second terminal, coupled to the capacitor and the third node; and a control terminal, configured to receive the control signal, wherein the third transistor is conducted in response to the control signal at the first stage; and a fourth transistor, comprising: a first terminal, coupled to the capacitor; a second terminal, configured to receive the data voltage from a data line; and a control terminal, configured to receive the control signal, wherein the fourth transistor is conducted in response to the control signal at the first stage.
3. The pixel circuit of claim 1, wherein the driving transistor comprising: a first terminal, coupled to the first node; a second terminal, coupled to the first transistor; and a control terminal, coupled to the second node, wherein the driving transistor is conducted in response to a voltage level of the second node.
4. The pixel circuit of claim 1, further comprising: a bypass transistor, comprising: a first terminal; a second terminal, coupled to the light emitting element; and a control terminal, configured to receive the driving signal, wherein the bypass transistor is conducted in response to the driving transistor at the second stage; and a detecting transistor, comprising: a first terminal; a second terminal, coupled to the first terminal of the bypass transistor; and a control terminal, configured to receive a detection control signal, wherein the detecting transistor is conducted in response to the detection control signal at a detecting stage.
5. A pixel circuit, comprising: a first light emitting element, coupled to a first system voltage source; a driving transistor, coupled to a first node, a second node and a second system voltage source; a capacitor, coupled to the second node and a third node; a writing circuit, coupled to the second node and the third node, wherein the writing circuit is configured to write an initial voltage and a first data voltage to both terminals of the capacitor respectively according to a control signal at a first stage terminal; a first driving circuit, coupled to the first node and the third node, wherein the first driving circuit is conducted in response to a first driving signal at a second stage to change the initial voltage and the first data voltage at both terminals of the capacitor at the first stage, so as to conduct the driving transistor, to light up the first light emitting element; a second light emitting element, coupled to the first system voltage source; and a second driving circuit, coupled to the first node and the third node, wherein the writing circuit is configured to write the initial voltage and a

second data voltage to both terminals of the capacitor according to the control signal at a third stage, wherein the second driving circuit is conducted in response to a second driving signal at a fourth stage to change the initial voltage and the second data voltage at both terminals of the capacitor at the third stage so as to conduct the driving transistor to light up the second light emitting element.

6. The pixel circuit of claim 5, wherein the writing circuit is configured to write the initial voltage and a third data voltage respectively according to the control signal at a fifth stage, wherein the pixel circuit further comprises: a third light emitting element, coupled to the first system voltage source; and a third driving circuit, coupled to the first node and the third node, wherein the third driving circuit is conducted in response to a third driving signal at a sixth stage to change the initial voltage and the third data voltage at both terminals of the capacitor at the fifth stage so as to conduct the driving transistor to light up the third light emitting element.

7. The pixel circuit of claim 6, wherein an optical wavelength of each of the first light emitting element, the second light emitting element and the third light emitting element is different.

8. The pixel circuit of claim 6, wherein the driving transistor is conducted in response to a voltage level of the second node at the second stage so that a first driving current is generated between the first system voltage source and the second system voltage source to flow through the driving transistor and a first transistor of the first driving circuit, thereby lighting the first light emitting element, wherein the driving transistor is conducted in response to a voltage level of the second node at the fourth stage so that a second driving current is generated between the first system voltage source and the second system voltage source to flow through the driving transistor and a second transistor of the second driving circuit, thereby lighting the second light emitting element, wherein the driving transistor is conducted in response to a voltage level of the second node at the sixth stage so that a third driving current is generated between the first system voltage source and the second system voltage source to flow through the driving transistor and a third transistor of the third driving circuit, thereby lighting the third light emitting element.

9. The pixel circuit of claim 8, wherein the first transistor is coupled between the driving transistor and the first light emitting element, wherein the second transistor is coupled between the driving transistor and the second light emitting element, wherein the third transistor is coupled between the driving transistor and the third light emitting element.

10. The pixel circuit of claim 9, further comprising: a detecting transistor, comprising: a first terminal; a second terminal; and a control terminal, configured to receive a detection control signal, wherein the detecting transistor is conducted in response to the detection control signal at a first detecting stage; a first bypass transistor, comprising: a first terminal, coupled to the second terminal of the detecting transistor; a second terminal, coupled to the first light emitting element; and a control terminal, configured to receive the first driving signal, wherein the first bypass transistor is conducted in response to the first driving signal at the second stage; a second bypass transistor, comprising: a first terminal, coupled to the second terminal of the detecting transistor; a second terminal, coupled to the second light emitting element; and a control terminal, configured to receive the second driving signal, wherein the second bypass transistor is conducted in response to the second driving signal at the fourth stage; and a third bypass transistor, comprising: a first terminal, coupled to the second terminal of the detecting transistor; a second terminal, coupled to the third light emitting element; and a control terminal, configured to receive the third driving signal, wherein the third bypass transistor is conducted in response to the third driving signal at the sixth stage.

11. The pixel circuit of claim 6, wherein an optical wavelength of each of the first light emitting element, the second light emitting element and the third light emitting element is different.

12. The pixel circuit of claim 6, wherein the first data voltage, the second data voltage and the third data voltage are different from each other.

13. The pixel circuit of claim 8, wherein a current path of each of the first drive current, the second

drive current and the third drive current are different.

14. The pixel circuit of claim 9, wherein the first transistor and the first bypass transistor are connected in series, wherein the second transistor and the second bypass transistor are connected in series, wherein the third transistor and the third bypass transistor are connected in series.

15. The pixel circuit of claim 10, wherein the first bypass transistor, the second bypass transistor and the third bypass transistor are connected in parallel.
